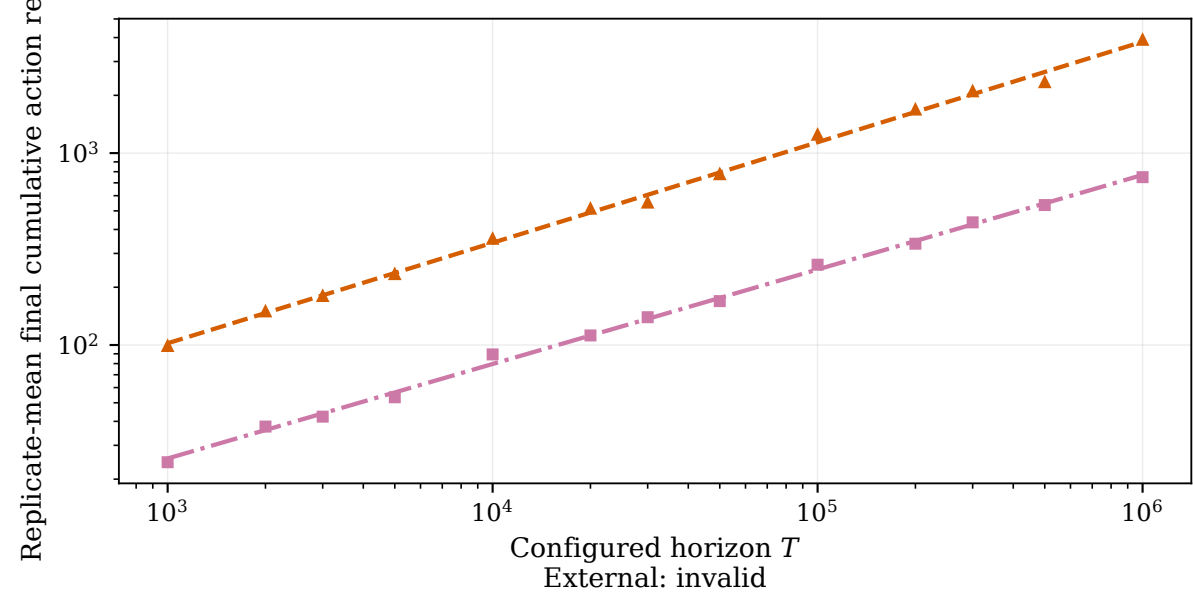


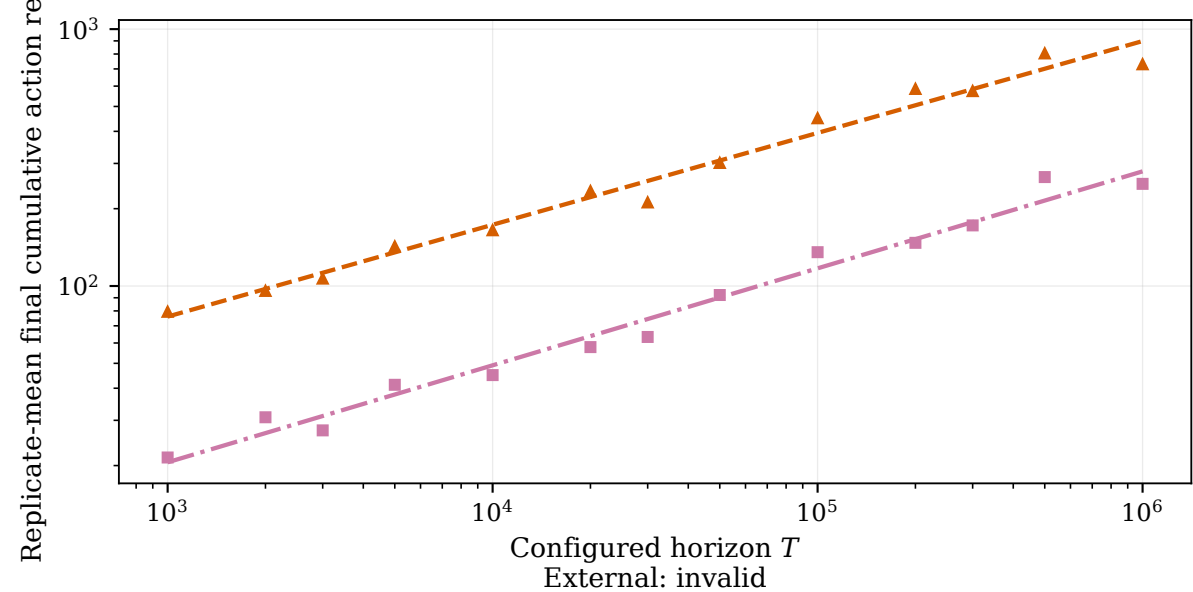
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: BM-Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.000175083, SD = 0.000465202, SE = 0.000134292
Internal R/T: mean = 0.000749333, SD = 0.000111192, SE = 3.20984e-05
Swap R/T: mean = 0.00389417, SD = 0.000468079, SE = 0.000135123
External: invalid
Internal: $\alpha = 0.493$, $c = 0.8531$, $R^2 = 0.998$
Swap: $\alpha = 0.524$, $c = 2.739$, $R^2 = 0.998$



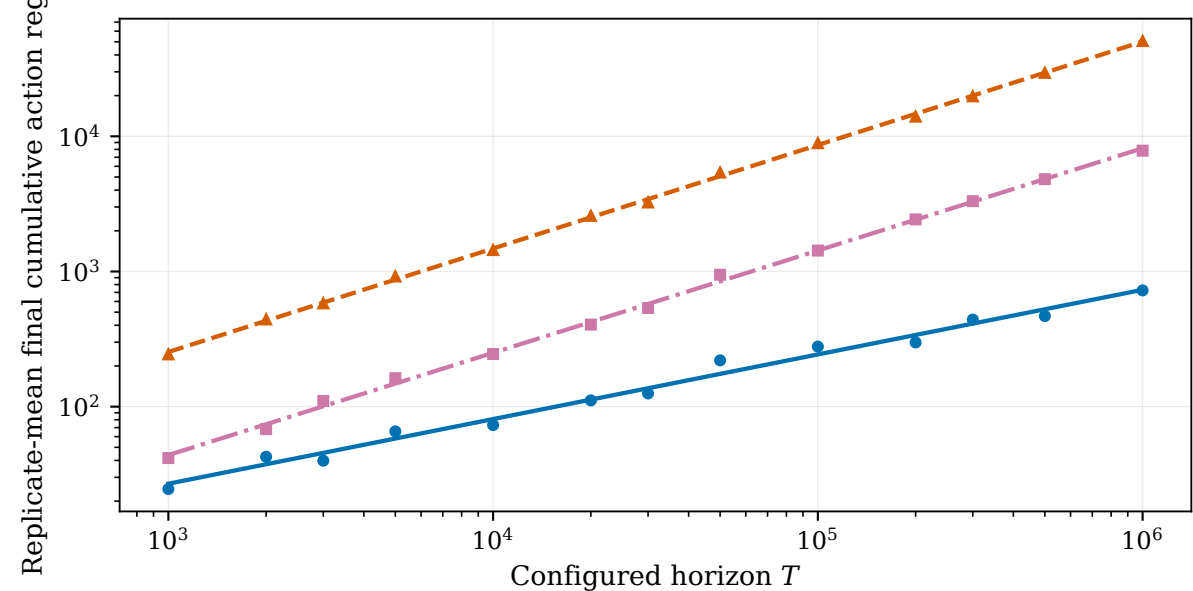
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: BM-OptHedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.00249917, SD = 0.00047666, SE = 0.0001376
Internal R/T: mean = 0.000249833, SD = 8.82711e-05, SE = 2.54817e-05
Swap R/T: mean = 0.000729333, SD = 0.000297851, SE = 8.59821e-05
External: invalid
Internal: $\alpha = 0.378$, $c = 1.518$, $R^2 = 0.980$
Swap: $\alpha = 0.357$, $c = 6.434$, $R^2 = 0.980$



Experiment Information

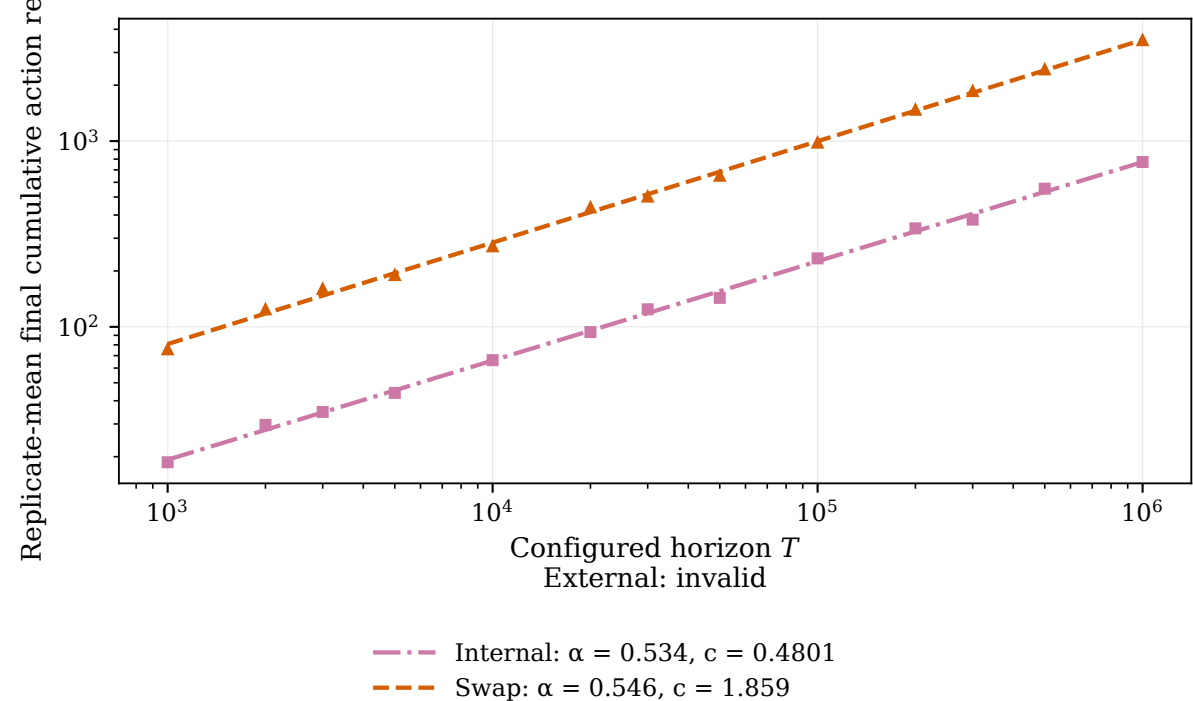
Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.0007245, SD = 0.000326643, SE = 9.42938e-05
Internal R/T: mean = 0.00783042, SD = 0.00149371, SE = 0.000431196
Swap R/T: mean = 0.0506502, SD = 0.00808444, SE = 0.00233378
External: $\alpha = 0.478$, $c = 0.9889$, $R^2 = 0.987$
Internal: $\alpha = 0.757$, $c = 0.2354$, $R^2 = 0.999$
Swap: $\alpha = 0.766$, $c = 1.282$, $R^2 = 1.000$



- External: $\alpha = 0.478, c = 0.9889$
- Internal: $\alpha = 0.757, c = 0.2354$
- Swap: $\alpha = 0.766, c = 1.282$

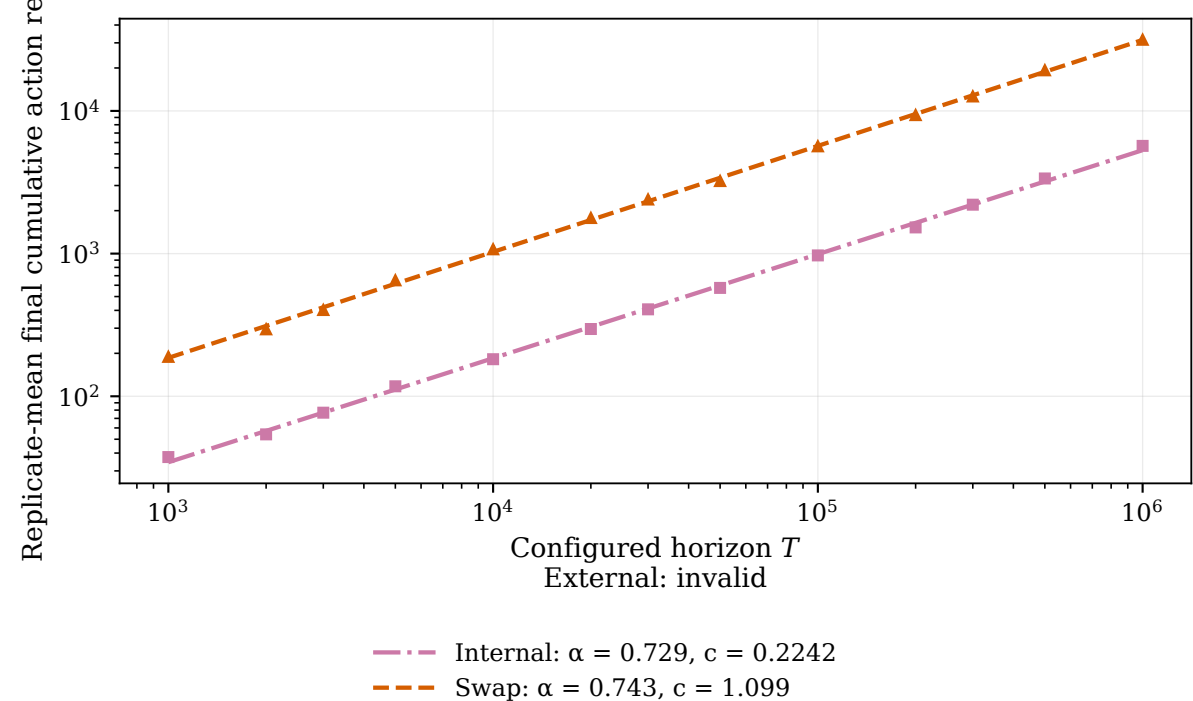
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: Ito-Hedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.000669917, SD = 0.00050081, SE = 0.000144571
Internal R/T: mean = 0.000771583, SD = 0.000209516, SE = 6.0482e-05
Swap R/T: mean = 0.00349483, SD = 0.000596753, SE = 0.000172268
External: invalid
Internal: $\alpha = 0.534$, $c = 0.4801$, $R^2 = 0.999$
Swap: $\alpha = 0.546$, $c = 1.859$, $R^2 = 0.999$



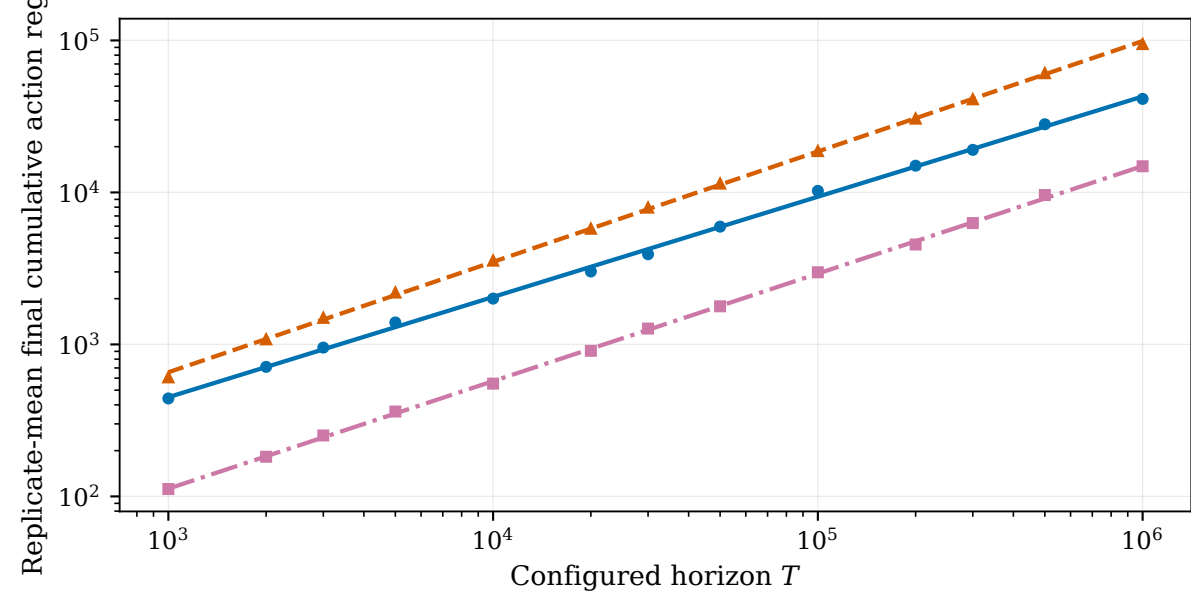
Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: OptHedge
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.00274242, SD = 0.00049422, SE = 0.000142669
Internal R/T: mean = 0.00568767, SD = 0.000769671, SE = 0.000222185
Swap R/T: mean = 0.0315117, SD = 0.00529378, SE = 0.00152818
External: invalid
Internal: $\alpha = 0.729$, $c = 0.2242$, $R^2 = 0.999$
Swap: $\alpha = 0.743$, $c = 1.099$, $R^2 = 1.000$



Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: RM
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.0412372, SD = 0.0104722, SE = 0.00302308
Internal R/T: mean = 0.014864, SD = 0.00232601, SE = 0.000671462
Swap R/T: mean = 0.0945782, SD = 0.00788139, SE = 0.00227516
External: $\alpha = 0.659$, $c = 4.725$, $R^2 = 0.999$
Internal: $\alpha = 0.708$, $c = 0.8469$, $R^2 = 1.000$
Swap: $\alpha = 0.726$, $c = 4.342$, $R^2 = 1.000$



- External: $\alpha = 0.659$, $c = 4.725$
- Internal: $\alpha = 0.708$, $c = 0.8469$
- Swap: $\alpha = 0.726$, $c = 4.342$

Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithm: SRM
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = -0.000268167, SD = 0.000794326, SE = 0.000229302
Internal R/T: mean = 0.000497833, SD = 8.72695e-05, SE = 2.51925e-05
Swap R/T: mean = 0.00301133, SD = 0.000492072, SE = 0.000142049
External: invalid
Internal: $\alpha = 0.490$, $c = 0.5236$, $R^2 = 0.995$
Swap: $\alpha = 0.494$, $c = 3.024$, $R^2 = 0.999$

